



```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  #define EMPTY -1
5
6  int hash(int key, int m) {
7      return key % m;
8  }
9
10 int main() {
11     int m, n, i, key, index, startIndex;
12
13     printf("Enter number of memory locations (m): ");
14     scanf("%d", &m);
15
16     int hashTable[m];
17
18     for (i = 0; i < m; i++) {
19         hashTable[i] = EMPTY;
20     }
21
22     printf("Enter number of employee records (N): ");
23     scanf("%d", &n);
24
25     for (i = 0; i < n; i++) {
26         printf("\nEnter 4-digit employee key: ");
27         scanf("%d", &key);
28
29         index = hash(key, m);
30         startIndex = index;
31
32         while (hashTable[index] != EMPTY) {
33             index = (index + 1) % m;
34
35             if (index == startIndex) {
36                 printf("Hash Table is full. Cannot insert key %d\n", key);
37                 break;
38             }
39     }
```



Start here x Lab10.c x

```
19     hashTable[i] = EMPTY;
20 }
21
22 printf("Enter number of employee records (N): ");
23 scanf("%d", &n);
24
25 for (i = 0; i < n; i++) {
26     printf("\nEnter 4-digit employee key: ");
27     scanf("%d", &key);
28
29     index = hash(key, m);
30     startIndex = index;
31
32     while (hashTable[index] != EMPTY) {
33         index = (index + 1) % m;
34
35         if (index == startIndex) {
36             printf("Hash Table is full. Cannot insert key %d\n", key);
37             break;
38         }
39     }
40
41     if (hashTable[index] == EMPTY) {
42         hashTable[index] = key;
43         printf("Key %d inserted at address %d\n", key, index);
44     }
45 }
46
47 printf("\nHash Table Contents:\n");
48 for (i = 0; i < m; i++) {
49     if (hashTable[i] == EMPTY)
50         printf("Address %02d : EMPTY\n", i);
51     else
52         printf("Address %02d : %d\n", i, hashTable[i]);
53 }
54
55 return 0;
56 }
57 }
```

Enter number of memory locations (m): 8
Enter number of employee records (N): 6

Enter 4-digit employee key: 2345
Key 2345 inserted at address 1

Enter 4-digit employee key: 2341
Key 2341 inserted at address 5

Enter 4-digit employee key: 7834
Key 7834 inserted at address 2

Enter 4-digit employee key: 9876
Key 9876 inserted at address 4

Enter 4-digit employee key: 5642
Key 5642 inserted at address 3

Enter 4-digit employee key: 7645
Key 7645 inserted at address 6

Hash Table Contents:

Address 00 : EMPTY
Address 01 : 2345
Address 02 : 7834
Address 03 : 5642
Address 04 : 9876
Address 05 : 2341
Address 06 : 7645
Address 07 : EMPTY

Process returned 0 (0x0) execution time : 49.034 s

Press any key to continue.